

MATH141 - Foundations of Engineering Mathematics

Answers to Selected Supplementary Exercises

Strand 1 - Fundamentals

1. (a) 2 (b) $(ag)^x$ (c) $b^x - b^y$
(d) 2^9 (e) $a^{-5}c^4$ (f) x^{3a}
2. (a) 81 (b) 3 (c) $\frac{1}{81}$
(d) $\frac{1}{125}$ (e) $\frac{8}{27}$ (f) 0.0016
(g) $8x^{-12}$ (h) $\frac{1}{4}$ (i) $\frac{x^3}{3}$
3. $b = x^{3/2}y^{-1}$
4. (a) 180 (b) $12\sqrt{15}$ (c) $\sqrt{5}$
(d) $\sqrt{3}$ (e) $3\sqrt{2}$ (f) $-5\sqrt{7}$
5. (a) $x = 32$ (b) $y = 54$
(c) $x = 45$ (d) $a = 11, b = -1$
(e) $c = 33$ (f) $a = 4, b = 2$
(g) $x = 46, y = -6$
6. (i) $\frac{\sqrt{7}}{7}$ (ii) $4\sqrt{3}$
(iii) $4(2 + \sqrt{3})$ (iv) $12\sqrt{2}$
(v) $14\sqrt{5}$ (vi) $10\sqrt{3} - 33\sqrt{2}$
7. (i) $\frac{\sqrt{7} - \sqrt{3}}{4}$ (ii) $\frac{5 - \sqrt{21}}{2}$
(iii) $3(\sqrt{5} + 2)$ (iv) $\frac{5 - 3\sqrt{2}}{7}$
8. (i) $\frac{1}{a^{1/2}}$ (ii) $a^{1/2}$ (iii) $\frac{1}{ax^2}$
(iv) $a^{1/12}$ (v) $\frac{a^{15}}{b^{10}}$
9. (i) 4 (ii) $\frac{1}{32}$ (iii) 25
10. $7, \frac{1}{27}, 5, 2, \frac{1}{4}$
11. (a) $-x$ (b) 0 (c) $\frac{1}{2}x$ (d) $11x$
12. (a) $2q$ (b) $p + q$
(c) $2p - q$ (d) 0
(e) $p + 2q + 1$ (f) $-(p + q + 1)$
13. (i) 1 (ii) $\frac{4}{5}$
14. (i) $y = e^{-2x}$ (ii) $y = cx$
(iii) $y = \frac{x}{e^x - 1}$ (iv) $y = \frac{E}{R}e^{-x/Rc}$
15. (i) $a(a - x)$
(ii) $x(3x^2 - x + 1)$
(iii) $a^2x^3(17x^2 + 2a^2)$
(iv) $(a + b)(a + c)$
(v) $(a + b)(5 + b)$
(vi) $(a + b)(2x + y)$
(vii) $(x - y)(3a - b)$
(viii) $(x^3 + 2)(x + 1)$
(ix) $(a + b)(ax + by + c)$
(x) $(x + 1)(x + 2)$
(xi) $(x + 3)^2$
(xii) $(x - 3)(x - 4)$
(xiii) $(x - 4)(x - 14)$
(xiv) $(x - 15y)(x - 5y)$
(xv) $(8 + x)(11 + x)$

- (xvi) $(x + 1)(x - 2)$
- (xvii) $(x + 2)(x - 3)$
- (xviii) $(x - 2)(x + 3)$
- (xix) $(x - 3)(x + 7)$
- (xx) $(x - 6a)(x + 7a)$
- (xxi) $(11 + x)(10 - x)$
- (xxii) $(x + 4)(2x - 7)$
- (xxiii) $(2x - 3)(3x + 1)$
- (xxiv) $(3x + 4)(x + 1)$
- (xxv) $(x + 2)(x - 2)$
- (xxvi) $(3 + 2a)(3 - 2a)$
- (xxvii) $(pq - 10a^2)(pq + 10a^2)$
- (xxviii) $(a + b + c)(a + b - c)$
- (xxix) $(1 + a - b)(1 - a + b)$
- (xxx) $y(2x + y)$
- (xxx i) $(x + y + a)(x + y - a)$
- (xxx ii) $(a - b + c + d)(a - b - c - d)$
- (xxx iii) $(x - y)(x^2 + xy + y^2)$
- (xxx iv) $(4 + y)(16 - 4y + y^2)$
- (xxx v) $(x + y)(x - y)(x^2 + xy + y^2)(x^2 - xy + y^2)$
- (xxx vi) $(a + b)(a^2 - ab + b^2 + 1)$
- (xxx vii) $xy(x + y)(x - y)^2$
- (xxx viii) $(a^2 + b^2)(a^4 - a^2b^2 + b^4)(a + b)(a - b)$
 $\times (a^2 + ab + b^2)(a^2 - ab + b^2)$
16. (a) $\frac{3y + 4x}{xy}$ (b) $\frac{75}{xy}$
(c) $\frac{y}{3x}$ (d) $\frac{y^2 - x^2 + x^2y^2}{x^2y^2}$
(e) $x - 1$ (f) $\frac{x^2 - y^2}{x^2 + y^2}$
(g) $\frac{-(x + 1)}{x^2(3 + x)}$ (h) $\frac{-x^2(2x + 3)}{2 + x}$
(i) $\frac{(a + b)(a - b)}{2}$
17. (a) $\frac{4(x + 1)}{5}$ (b) $\frac{a^4b^2 - b^4c^2 + a^2c^4}{a^2b^2c^2}$
(c) $\frac{2x + 5}{(x + 2)(x + 3)}$ (d) $\frac{x^2 + y^2}{xy(x^2 - y^2)}$
(e) $\frac{4(x - 1)}{(x + 2)(x - 2)^2}$ (f) $\frac{5x^2}{(5x + y)(5x - y)}$
(g) $\frac{2}{(x - 4)(x - 6)}$
(h) $\frac{1}{(x - 1)(2x + 1)(3x - 2)}$
(i) $\frac{x^2 + 11}{(x - 1)(x + 2)(x + 3)}$
(j) $\frac{2x + 13}{(x + 3)(x + 4)(x - 4)}$

- (k) 0
- (l) $\frac{a-c}{(a-b)(b-c)}$
18. (i) $x = \pm\sqrt{2}$ (ii) $x = \frac{2}{5}$ or $-\frac{3}{2}$
19. (i) $x = \pm\frac{3}{2}$ or ± 1
 (ii) $y = -64$ or 8
20. (i) real; $r_1 + r_2 = 6, r_1 r_2 = -11$
 (ii) not real; $r_1 + r_2 = -\frac{3}{2}, r_1 r_2 = \frac{5}{2}$
 (iii) real; $r_1 + r_2 = 3, r_1 r_2 = \frac{9}{4}$
 (iv) not real; $r_1 + r_2 = 0, r_1 r_2 = \frac{8}{5}$
21. $x + 2y = 5, -\frac{1}{2}, 5, \frac{5}{2}$
22. (a) $3x - 7y + 15 = 0$

- (b) $x - 4y + 17 = 0$
23. (a) $(1, 8)$ (b) $(\frac{29}{5}, -3)$
24. triangle is isosceles
28. (b) $\frac{\sqrt{2-\sqrt{2}}}{2}$ (d) $-\frac{24}{25}, \frac{7}{25}, -\frac{24}{7}$
 (e) $\frac{1}{2}(\cos 4\theta + \cos 2\theta), \frac{1}{2}(\sin 8\theta + \sin 2\theta)$
32. (a) $\sqrt{549}$ (b) $\sqrt{137 - 44\sqrt{2}}$
33. 7332.12 km
34. $a = 10(\sqrt{3} - 1), b = 5\sqrt{6}(\sqrt{3} - 1)$
35. noon ± 2.14 minutes
36. 7.5